

Contents

`nvim scripts/apex/water_if_else.apex`

Add

```
String waterLevel = 'full'; /*This variable keeps track of the water level status: full or empty*/
if (waterLevel == 'empty') {
    System.debug('Fill the tea kettle');
    waterLevel = 'full'; /*Once the tea kettle is filled the variable is changed to full*/
} else {
    System.debug('The tea kettle is full');
}
```

- Run

`sf apex run --file scripts/apex/water_if_else.apex --target-org trailheadPlayground`

and Look for a `USER_DEBUG` line saying “The tea kettle is full”

`nvim scripts/apex/water_if_elseif.apex`

Prerequisites Make sure the Salesforce CLI is installed and that you have already authenticated to your Trailhead Playground or dev org.

Recommended one-time checks:

```
bash sf org list sf config set target-org=trailheadPlayground mkdir -p
scripts/apex sf org list shows your authorized orgs, and setting target-org lets
later commands omit the org flag if you want a shorter workflow.
```

Workflow pattern Whenever Trailhead says:

Debug -> Open Execute Anonymous Window

Paste code

Check Open Log

Click Execute

View Debug Only

Use this local workflow instead:

Create a `.apex` file in `scripts/apex/`.

Paste the anonymous Apex snippet into that file.

Run it with `sf apex run --file --target-org trailheadPlayground`.

Read the `USER_DEBUG` lines in the terminal output.

While loop exercise Create the file:

`bash nvim scripts/apex/while_loop_teatime.apex` Paste this code:

```

//Declare an Integer variable called totalGuests and set it to 100
Integer totalGuests = 100;
//Declare an Integer variable called totalAmountSugar and set it to 159 (number of teaspoons)
Integer totalAmountSugar = 159;
//Declare an Integer variable called totalAmountTea and set it to 35 (number of teabags).
Integer totalAmountTea = 35;
//Loop: Add a teaspoon of sugar and one tea bag to a tea cup. Serve until there is no sugar
while (totalGuests > 0) {
    System.debug(totalGuests + ' Guests Remaining');
    //check ingredients
    if (totalAmountSugar == 0 || totalAmountTea == 0) {
        System.debug('Out of ingredients! Sugar: ' + totalAmountSugar + ' Tea: ' + totalAmountTea);
        break; //ends the while loop
    }
    //add sugar
    totalAmountSugar--;
    //add tea
    totalAmountTea--;
    //guest served
    totalGuests--;
}
```apex

```

Run it:

```
sf apex run --file scripts/apex/while_loop_teatime.apex --target-org trailheadPlayground
```

This demonstrates a while loop, which checks its condition before each iteration and may run zero times if the condition starts false.

Do-while loop exercise Create the file:

```
nvim scripts/apex/do_while_loop_teatime.apex
```

Paste this code:

```

//Declare an Integer variable called totalGuests and set it to 100
Integer totalGuests = 100;
//Declare an Integer variable called totalAmountSugar and set it to 159 (number of teaspoons)
Integer totalAmountSugar = 159;
//Declare an Integer variable called totalAmountTea and set it to 35 (number of teabags).
Integer totalAmountTea = 35;
do {
 //deduct sugar serving
 totalAmountSugar--;
 //deduct tea serving
 totalAmountTea--;
 //deduct guest
 totalGuests--;
} while (totalGuests > 0);

```

```

 System.debug(totalGuests + ' Guests Remaining!');
 //check ingredients
 if (totalAmountSugar == 0 || totalAmountTea == 0) {
 System.debug('Out of ingredients!');
 break; //ends the loop
 }
} while (totalGuests > 0);

```

Run it:

```
sf apex run --file scripts/apex/do_while_loop_teatime.apex --target-org trailheadPlayground
```

This demonstrates a do-while loop, which executes the body once before checking the condition.

Optional log commands If you want more than the immediate terminal output, Salesforce CLI also includes Apex log commands for listing and tailing logs.

Useful extras:

`bash sf apex tail log -color sf apex log list sf apex tail log streams` logs in the terminal, which is useful when you want a CLI-first version of watching debug output continuously.

Suggested Neovim flow A fast Neovim loop for this module is:

Open a new anonymous Apex file under `scripts/apex/`.

Paste the Trailhead snippet.

Save the file.

Run your Neovim mapping or terminal command for `sf apex run -file %`.

Read the `USER_DEBUG` output in the terminal split.

If you added a mapping like `:SfApexRun`, then each exercise becomes:

```
text :edit scripts/apex/while_loop_teatime.apex :write :SfApexRun Minimal
command list
```

```

mkdir -p scripts/apex
nvim scripts/apex/while_loop_teatime.apex
sf apex run --file scripts/apex/while_loop_teatime.apex --target-org trailheadPlayground
nvim scripts/apex/do_while_loop_teatime.apex
sf apex run --file scripts/apex/do_while_loop_teatime.apex --target-org trailheadPlayground

```